Linear Regression

Kaung Myat Khant

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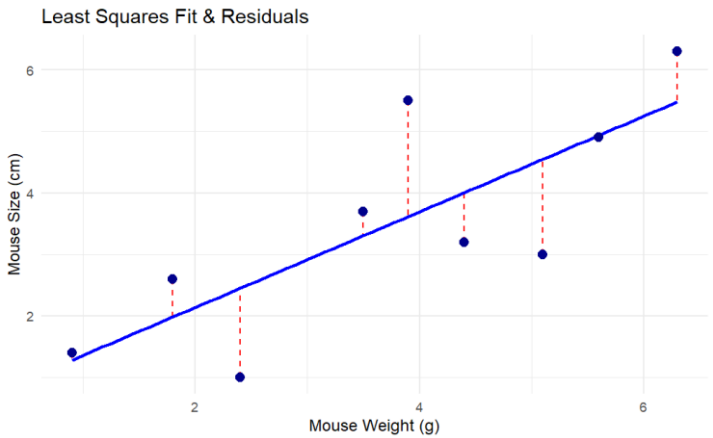
Three main concepts of Linear Regression

Linear regression quantifies the relationship between variables through three fundamental steps:

1. **Fit a line to the data** using **Least Squares**
2. **Calculate** R^2 to quantify how much variation is explained
3. **Calculate a** p**-value for** R^2 **or slope (beta)** (via the F-statistic) to ensure the relationship is statistically significant

   Both a **large** R^2 and a **small** p**-value** are required for a meaningful result.





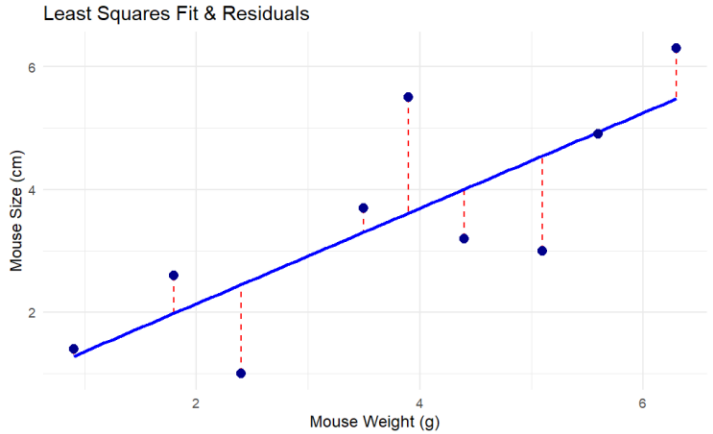
Example: Mouse Weight vs Size

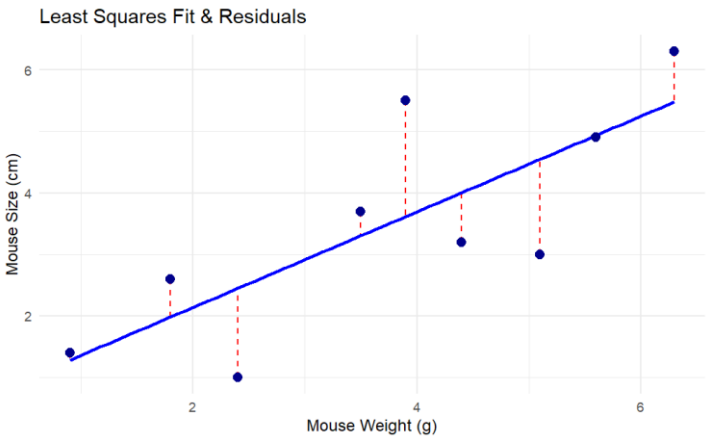
To make these concepts clear, we use a simple, concrete dataset:

- **Predictor variable (X)**: Mouse Weight
- **Outcome variable (Y)**: Mouse Size

The Least Squares Rotation Process

1. Draw an initial line through the dataset.
2. Measure the vertical distance (residual) from each point to the line.
3. Square each distance and sum them together.
4. Slightly rotate the line and re-calculate the sum of squared residuals.
5. Identify the rotation angle that yields the **minimum (least) sum of squared residuals**.

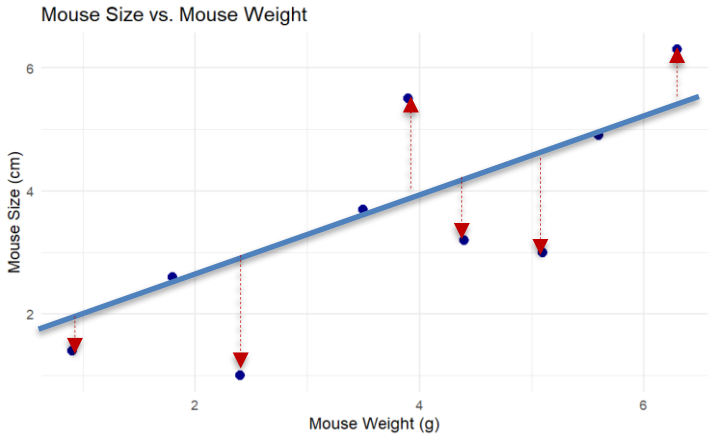




Example: Mouse Weight vs Size

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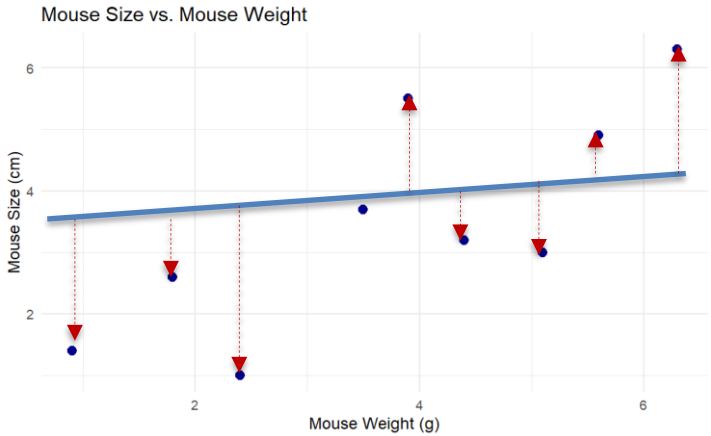
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Example: Mouse Weight vs Size

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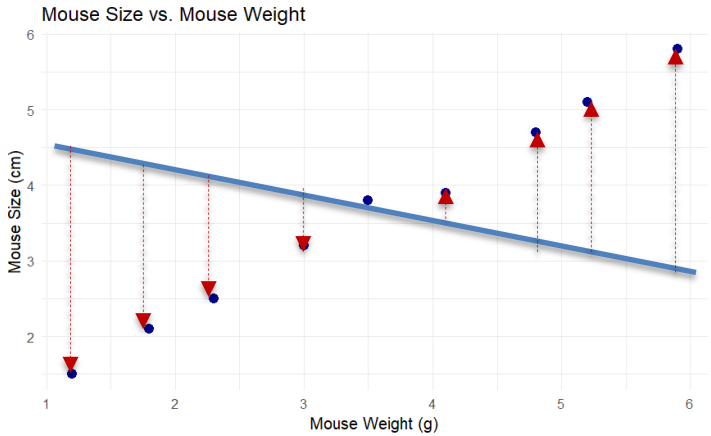
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Example: Mouse Weight vs Size

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Residuals and the Sum of Squared Residuals

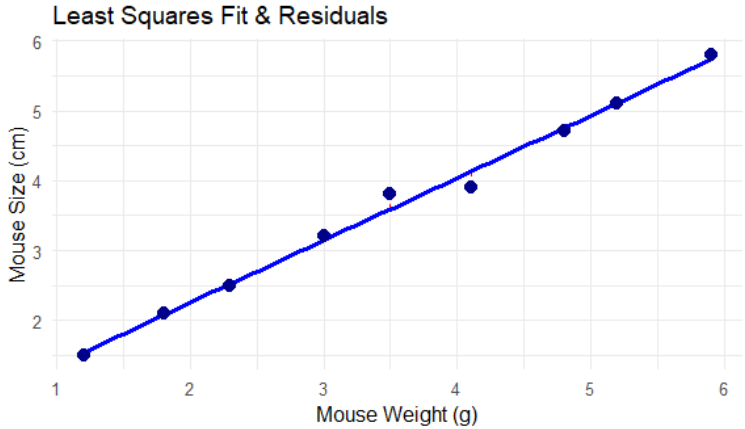
- **Residual**: The vertical distance from an observed data point to the candidate line.
  - ❖ Residual$_i = y_i - \hat{y}_i$
- **Sum of Squared Residuals (SS_{fit})**:
  - ❖ $SS_{\text{fit}} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

By squaring the residuals:

- Negative and positive distances don't cancel each other out.
- Larger errors are penalized more heavily.

The Least Squares Rotation Process

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The Fitted Line Equation

The least squares procedure estimates two key parameters:

$$y = a + b \cdot x$$

- a **(y-intercept)**: The baseline value of y when $x = 0$.
- b **(slope)**: The estimated change in y for every 1-unit increase in x.

A non-zero slope indicates that knowing the predictor (x) provides information about the outcome (y).

Quantifying Fit with R^2

Variation Around the Mean vs. Variation Around the Fit

To measure how well the line fits:

1. **Variation Around the Mean (SS_{mean})**:
   - Calculate mean of the outcome (y) (ignore predictor x).
   - Calculate variance around the average outcome $\bar{y}$:

$$SS_{\text{mean}} = \sum(y_i - \bar{y})^2, \quad \text{Var(mean)} = \frac{SS_{\text{mean}}}{n}$$

2. **Variation Around the Fit (SS_{fit})**:
   - Calculate variance around the fitted line:

$$SS_{\text{fit}} = \sum(y_i - \hat{y}_i)^2, \quad \text{Var(fit)} = \frac{SS_{\text{fit}}}{n}$$

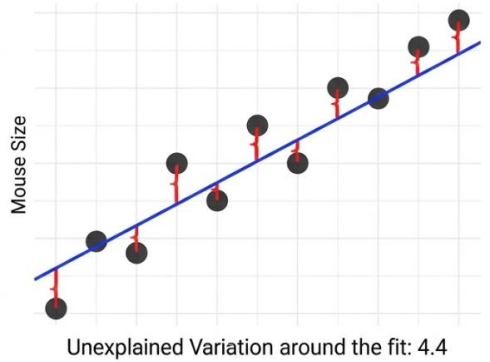
R^2 Definition & Formula

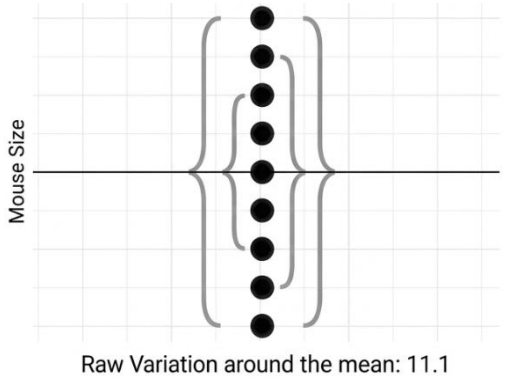
R^2 quantifies the **percentage reduction in variance** achieved by including the predictor:

$$R^2 = \frac{\text{Var(mean)} - \text{Var(fit)}}{\text{Var(mean)}} = \frac{SS_{\text{mean}} - SS_{\text{fit}}}{SS_{\text{mean}}}$$

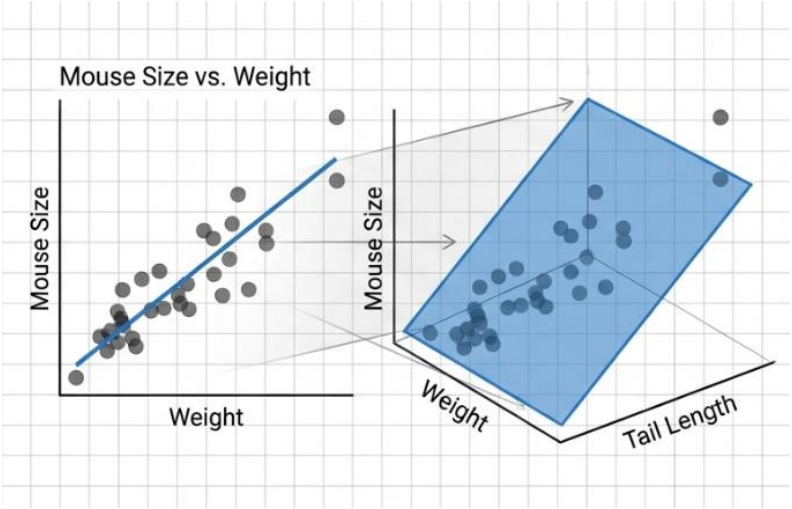
Interpreting R^2 Values:

- $R^2 = 1.0$ **(100%)**: Perfect fit; line passes through all points.
- $R^2 = 0.60$ **(60%)**: 60% of the variation in size is explained by weight.
- $R^2 = 0.0$ **(0%)**: Predictor provides no useful information.





$$R^2 = \frac{11.1 - 4.4}{11.1} = 0.604$$

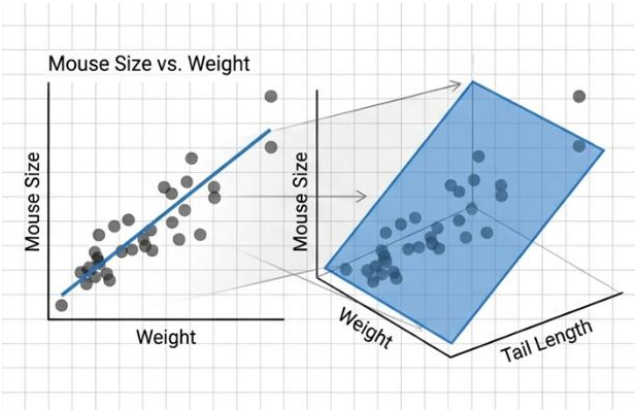




Mouse Weight
+ Tail Length
+ Coin Flip +
Astrological Sign

Multiple Regression & Adjusted R square

Multiple Regression & Adjusted R square



Expanding to Multiple Predictors

- Adding variables (e.g., body length predicted by weight AND tail length) expands the fit from a 2D line to a 3D plane (or hyperplane).

The Overfitting Problem

- Adding parameters will **never decrease** raw R^2, because least squares can zero out useless parameters.
- Random noise variables can artificially inflate R^2 by chance.



Mouse Weight
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Solution: Adjusted R^2

Penalizes R^2 based on the number of added parameters to prevent misleadingly high values from extra noise variables.

Assessing Significance with p-Values

Why R^2 Alone Is Not Enough

- Any 2 data points can be connected by a perfect straight line, yielding $R^2 = 1.0$ (100%).
- However, an $R^2 = 1.0$ from only 2 samples has no statistical reliability.
- We need a **p-value** to determine if the calculated R^2 could have occurred purely by random chance.

The F-Statistic

The p-value for R^2 is derived from the **F-statistic**, which compares explained variance to unexplained variance:

$$F = \frac{\text{Variation explained by predictor(s)}}{\text{Variation NOT explained by predictor(s)}}$$

Detailed Formula with Degrees of Freedom:

$$F = \frac{\frac{SS_{\text{mean}} - SS_{\text{fit}}}{p_{\text{fit}} - p_{\text{mean}}}}{\frac{SS_{\text{fit}}}{n - p_{\text{fit}}}}$$

Where:

- p_{fit}(k) = number of parameters in the fit model (e.g., 2 for intercept + slope)
- p_{mean}(1) = number of parameters in the mean model (1 for intercept)
- n = total sample size

F-Statistic Simulation & p-Value Calculation

1. **Simulation Thought Experiment (F distribution for Null)**:
   - Generate thousands of random datasets with no real relationship.
   - Calculate the F-statistic for each dataset and build an F-distribution.
2. **Calculating the p-Value**:
   - Locate the observed F-statistic on the distribution curve.
   - The p-value is the proportion of random outcomes as extreme as or more extreme than the observed F.

     More sample points (n) relative to parameters (p_{fit}) steepen the curve, yielding smaller p-values for the same effect size.

Core Rule of Thumb:

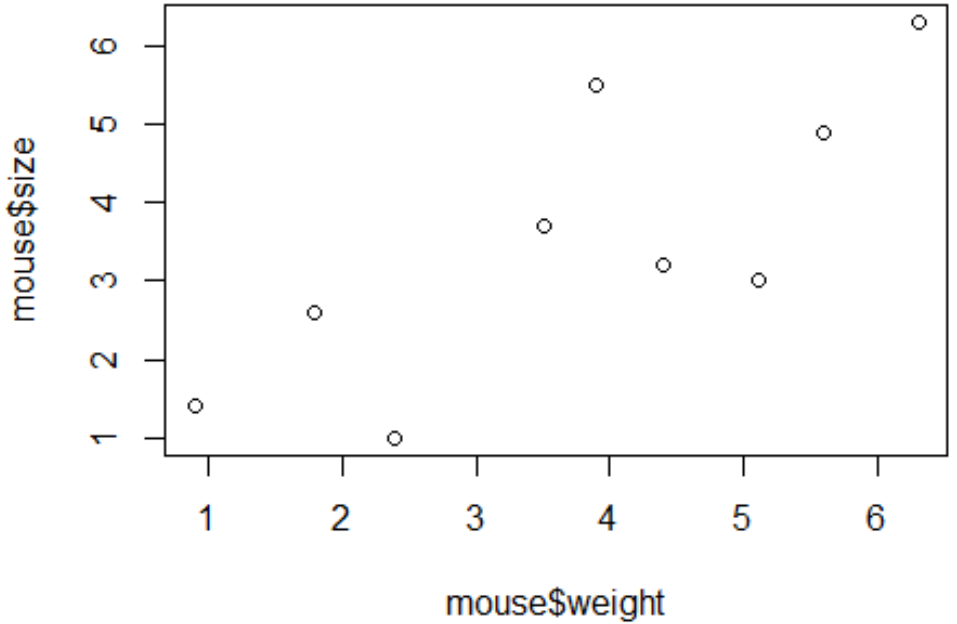
An interesting and reliable relationship requires **both a large R^2 AND a small p-value**.

R examples (Data)

```
##    size weight tail
## 1   1.4    0.9  0.7
## 2   2.6    1.8  1.3
## 3   1.0    2.4  0.7
## 4   3.7    3.5  2.0
## 5   5.5    3.9  3.6
## 6   3.2    4.4  3.0
## 7   3.0    5.1  2.9
## 8   4.9    5.6  3.9
## 9   6.3    6.3  4.0
```

R examples - Plotting simple linear regression

```
plot(mouse$weight, mouse$size)
```

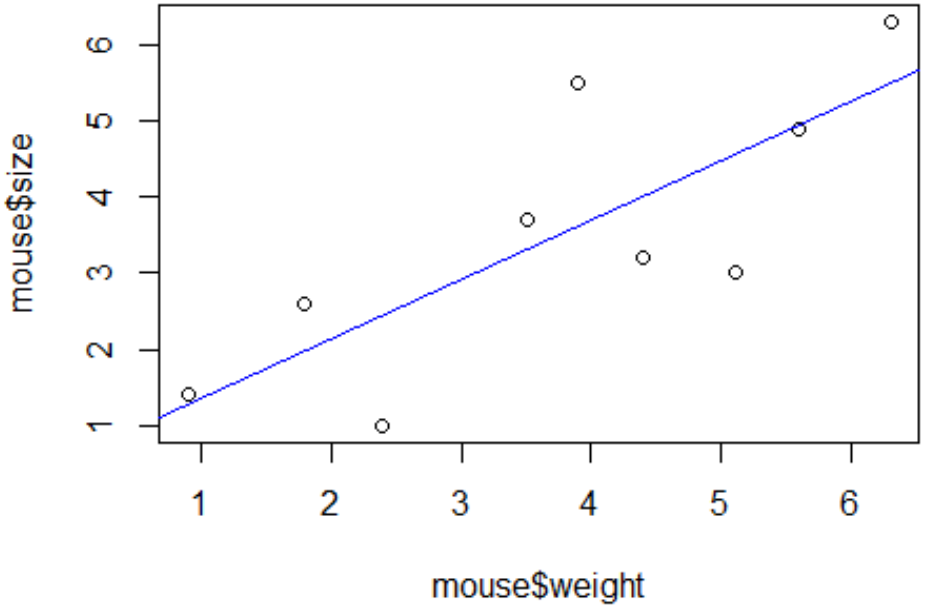


R examples - Simple linear regression

```
mouse_lm <- lm(formula = size ~ weight, data = mouse)
summary(mouse_lm)
##
## Call:
## lm(formula = size ~ weight, data = mouse)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.5482 -0.8037  0.1186  0.6186  1.8852
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.5813     0.9647   0.603   0.5658
## weight        0.7778     0.2334   3.332   0.0126 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.19 on 7 degrees of freedom
## Multiple R-squared:  0.6133, Adjusted R-squared:  0.558
## F-statistic:  11.1 on 1 and 7 DF,  p-value: 0.01256
```

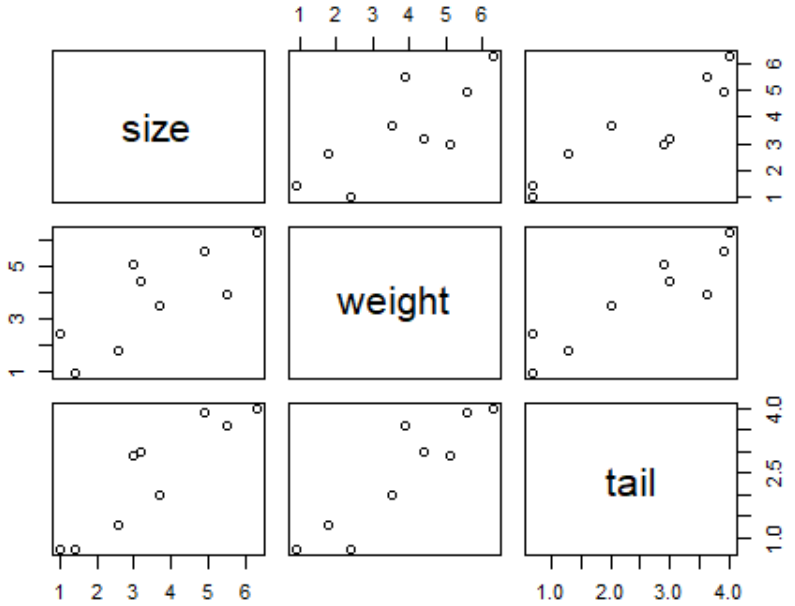
R examples - Smoothing simple linear regression

```
plot(mouse$weight, mouse$size)
abline(mouse_lm, col = "blue")
```



R examples - Plotting multiple linear regression

```
plot(mouse)
```



R examples - Multiple linear regression

```
mouse_lm_multi <- lm(formula = size ~ weight + tail, data = mouse)
summary(mouse_lm_multi)
##
## Call:
## lm(formula = size ~ weight + tail, data = mouse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.99928 -0.38648 -0.06967  0.34454  1.07932
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.7070     0.6510   1.086   0.3192
## weight       -0.3293     0.3933  -0.837   0.4345
## tail          1.6470     0.5363   3.071   0.0219 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8017 on 6 degrees of freedom
## Multiple R-squared:  0.8496, Adjusted R-squared:  0.7995
## F-statistic: 16.95 on 2 and 6 DF,  p-value: 0.003399
```

Thanks

Reference

1. Starmer J.; ***Linear Regression and Linear Models***; Stat Quest; University of North Carolina